



Trabajo Práctico N°5 B

Cinemática Inversa

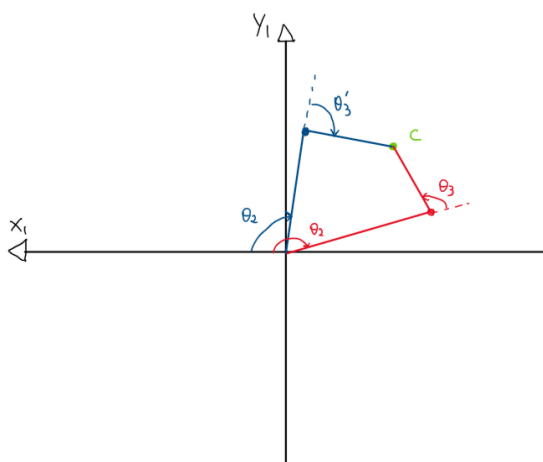
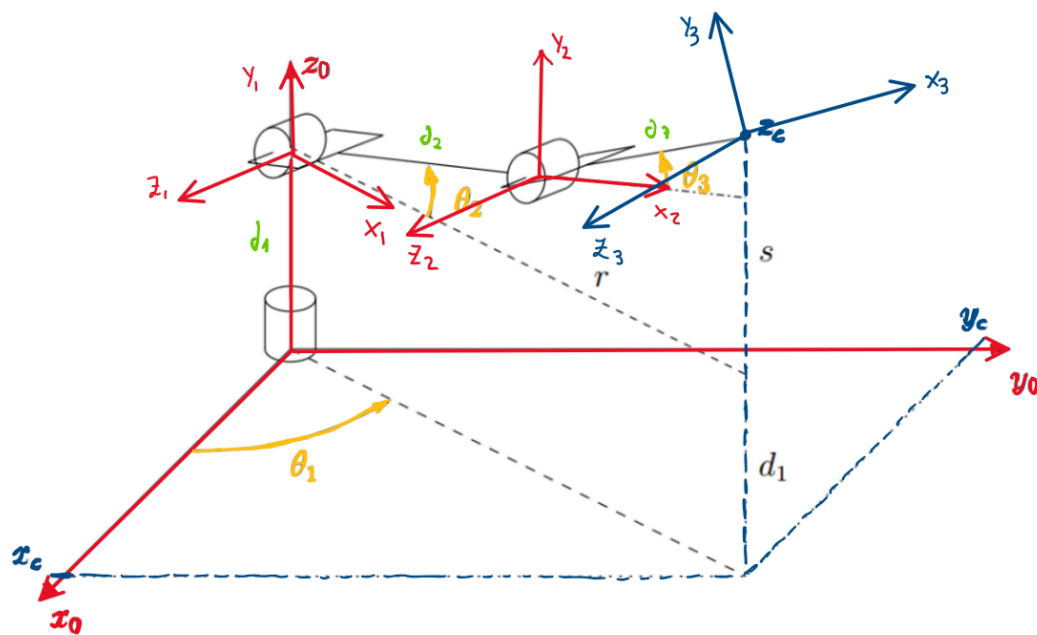
Grupo: 7

Integrantes:

- Bartolini Ignacio - 12911
- Francisco Castel -13784
- Joaquín Calderón - 13839

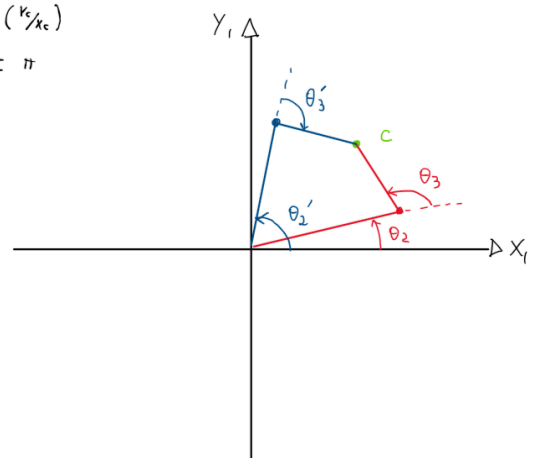
Ingeniería en Mecatrónica

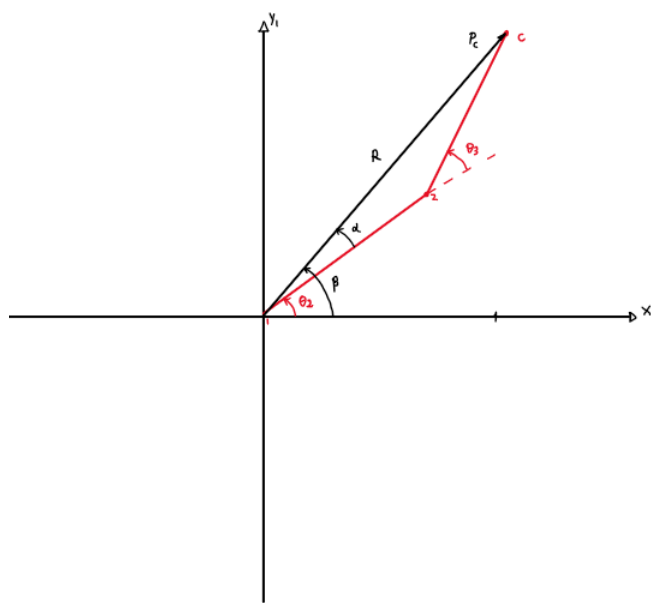
Ejercicio 1



$$\theta_1 = \arctan 2 (y_c/x_c)$$

$$\theta_1' = \theta_1 \pm \pi$$





$$r = \sqrt{x_c^2 + y_c^2}$$

$$\beta = \arctan 2 \left(\frac{y_c}{x_c} \right) \quad * \begin{matrix} 5 & 2 \\ 2 & 5 \end{matrix}$$

$$\cos \theta_3 = \frac{r^2 - d_2^2 - d_3^2}{2 d_2 \cdot d_3}$$

$$\rightarrow \pm \theta_3 = \pm \arccos \left(\frac{r^2 - d_2^2 - d_3^2}{2 d_2 \cdot d_3} \right)$$

$$\theta_2 = \beta - \arctan 2 \left(\frac{d_3 \sin \theta_3}{d_2 + d_3 \cos \theta_3} \right)$$

Ejercicio 2